

```

•         step = 0.1
•
•         duty = mapFun(sin(x),-1,1,0,100) # Call our special
map function
•         x += step                                # Equivalent to:
x = x + step
•         pwm.ChangeDutyCycle(duty)
•         print(duty)
•         # If you want to remove the print, make sure you
have a time.sleep
•         # Otherwise the program will run too fast and
spoil the effect.
•         #time.sleep(0.01)
•         if x >= 2*math.pi:                        # Keep x bounded
(0 to 2Pi)
•         x = 0
•     except KeyboardInterrupt:
•         pwm.stop()
•         GPIO.cleanup()
•         print('Halted Cleanly')

```

File Permissions Booting And Everything Else

File permissions and information for all other systems is obtained by sending arguments to the mainframe. These are nothing but pieces of information that are passed into a program or function. When you issue a command like `cd /home/pi`, here the command is `cd` with one argument, `/home/pi`.

Booting On Linux System

```

• #!/bin/bash
• # Archive the input file to a predetermined location
• # Check we have exactly 1 argument
• if [ $# -ne 1 ]
• then
•     echo "Usage: $0 FILE"
•     exit 1 # Quit the script, indicate an error to the
shell
• fi
• # The archive location
• archive=/home/pi/archives
•
• # Make the destination if it does not already exist

```